

TECHNICAL DEVELOPMENT REPORT

Baby Soothing Rocker Device

Phase 1 & Phase 2 — Progress Documentation

Date: June 8, 2026

1. Project Overview

This document records all development work completed on the Baby Soothing Rocker Device project from initial setup through motor testing, Vtech reverse engineering, MCU evaluation, and IMU integration. It serves as an internal technical reference and progress log.

1.1 Objective

The End goal is to design and prototype a vibration-based baby soothing rocker device with eccentric mass motor mechanism, BLE app control, USB MSC audio file management, IMU vibration feedback, cry detection, and adjustable speed levels — comparable to Vtech V-Hush commercial devices.

1.2 Development Platform

Item	Detail
Primary MCU (finalized)	ESP32-S3-N16R8
Development MCU (initial)	STM32F103C8T6 (Blue Pill clone)
Programmer	ST-Link from STM32F429-DISC1
Motor Driver	TB6612FNG
IDE	STM32CubeIDE, ESP-IDF, Arduino IDE
Target Voltage	3.7 V Li-ion (3.6–4.2 V)

2. Development Sprint Log

All work completed to date is summarised below in chronological sprint order.

#	Sprint	Description / Outcome	Status
1	STM32 Initial Setup	Installed STM32CubeIDE, configured workspace, created first test program	Done
2	STM32F429-DISC1 Inspection	Confirmed Discovery board functional; identified Blue Pill as non-working via SWD	Done
3	Blue Pill Recovery	Replaced faulty board, resolved NRST/Connect-Under-Reset issues, confirmed SWD working	Done
4	UART Debug Setup	Configured USART for serial debug output; verified with terminal	Done
5	TB6612 Soldering	Soldered headers onto TB6612 motor driver module	Done

6	Motor Driver Code	Wrote PWM control code for TB6612; tested direction and speed	Done
7	Vibration Motor Test	Tested vibration motor across varying voltages; recorded response	Done
8	Mechanism Study	Studied ERM motion, crank-rocker and pendulum linkage mechanisms	Done
9	IMU Sourcing	Sourced ICM-42688 and ICM-45686 from China suppliers	Done
10	Multi-Motor Testing	Tested M2–M7 at different voltages and load conditions with TB6612	Done
11	Vtech Disassembly	Disassembled Vtech V-Hush Baby Rocker; documented PCB and components	Done
12	Vtech BOM & Functional Test	Compiled full BOM; tested speaker, LEDs, charger IC, boost IC, motor individually	Done
13	Vtech Reassembly	Reassembled and verified full Vtech unit operational after testing	Done
14	Motor Order Finalization	Evaluated all tested motors; placed order for finalized motor selection	Done
15	Vibration Mechanism CAD	Designed eccentric weight mount CAD for motor shaft	Done
16	Sling Mechanism Study	Studied sling-based soothing motion; evaluated ERM vs pendulum approach	Done
17	CAD Finalization	Finalized mechanical CAD design for vibration mount mechanism	Done
18	New Motor Testing	Tested newer motors with data recording; finalized motor selection	Done
19	MCU Evaluation	Compared RP2040, STM32F103, ESP32-S3 across USB MSC, BLE, WiFi, timers, app control	Done
20	MCU Compatibility Check	Ran RP2040 and STM32 to verify peripheral compatibility in prototype	Done
21	Motor + ESP32-S3 Integration	Soldered ESP32-S3 headers, set up ESP-IDF, tested motor with TB6612 at 3 speed levels	Done
22	ESP32-S3 Motor Test	Observed L1 too slow with noise; L2 good; L3 too much — tuning needed	Done
23	Vibration Meter Research	Evaluated GM63A, AS63B, Lutron VB8206SD; compared vs MPU6050 Python setup	Done
24	IMU Integration	Integrated MPU6050 with ESP32-S3; tuned filter constants for vibration sensitivity	Done

3. STM32 Development Setup

3.1 Environment Setup

- Installed STM32CubeIDE with STM32CubeMX integration.
- Configured workspace, created first GPIO LED blink test project targeting STM32F103C8T6.
- Used ST-Link extracted from STM32F429-DISC1 Discovery board by removing CN3 jumpers.
- Connected Blue Pill via CN2 SWD header: SWDIO (Pin 4), SWCLK (Pin 2), GND (Pin 3), NRST (Pin 5).

3.2 Blue Pill Troubleshooting

Initial Blue Pill board failed to enumerate on SWD. The following issues were identified and resolved:

Issue	Resolution
No device found on target	CN3 jumpers not removed on Discovery board — removed both
Target held under reset	NRST wire was shorting to GND — wiring corrected
SWD disconnects after flash	Connect Under Reset disabled after flashing; Software Reset used for reconnection
Clone chip UNEXPECTED idcode	GD32/CKS32 clone — reduced SWD frequency to 1000 kHz, ignored warning

3.3 LED Blink Test

Onboard LED confirmed on GPIOB. Test code using HAL_GPIO_TogglePin and HAL_Delay(500) compiled and ran successfully on the replacement Blue Pill.

4. Motor Driver — TB6612FNG

4.1 Hardware Preparation

- Soldered 0.1-inch headers onto bare TB6612 module.
- Wired AIN1/AIN2 for direction control, PWMA for speed, STBY pulled high.
- Powered motor supply from bench PSU at 6 V; logic from 3.3 V STM32 pin.

4.2 Firmware

Initial motor driver code written for STM32 using HAL TIM PWM output on TIM2 Channel 1. Three speed levels defined:

Level	PWM Duty	Board Voltage
L1	33%	4.5 V
L2	66%	5.0 V
L3	100%	6.0 V

Same code was later ported to ESP32-S3 using ESP-IDF LEDC peripheral with identical speed level definitions.

5. Motor Testing Results

5.1 Test Setup

- All motors tested at 6 V supply via TB6612 driver.
- Current measured with inline multimeter at each PWM speed level.
- Load condition tested with eccentric mass attached to motor shaft.
- Observations compared subjectively against Vtech device vibration feel.

5.2 Measurement Data

Section	Motor	Condition	Volt (V)	Level	Speed	Current (A)	Power (W)	Observation
Board Reference	Board	—	—	L1	—	—	—	4.5 V on board
Board Reference	Board	—	—	L2	—	—	—	5.0 V on board
Board Reference	Board	—	—	L3	—	—	—	6.0 V on board
Vtech Motor – PS Test	Vtech	PS Test	3.3	—	—	—	—	Works normal
Vtech Motor – PS Test	Vtech	PS Test	4.0	—	—	—	2.0	Max — OK
Vtech Motor – PS Test	Vtech	PS Test	6.0	—	—	—	—	Too much
M8 – No Load	M8	No load	6	L1	33%	0.35	2.10	—
M8 – No Load	M8	No load	6	L2	66%	0.50	3.00	—
M8 – No Load	M8	No load	6	L3	100%	0.42	2.53	—
M8 – With Load	M8	With load	6	L1	33%	0.36	2.16	Normal vibrations

M8 – With Load	M8	With load	6	L2	66%	0.66	3.96	~Vtech L3 level
M8 – With Load	M8	With load	6	L3	100%	1.10	6.60	Too much vibration
M7 – No Load	M7	No load	6	L1	33%	0.049	0.29	—
M7 – No Load	M7	No load	6	L2	66%	0.070	0.42	—
M7 – No Load	M7	No load	6	L3	100%	0.076	0.46	—
M7 – With Load	M7	With load	6	L1	33%	0.044	0.26	Too slow
M7 – With Load	M7	With load	6	L2	66%	0.077	0.46	Too slow
M7 – With Load	M7	With load	6	L3	100%	0.082	0.43	Still too slow
M1 – With Load	M1	With load	6	L1	33%	0.30	2.00	Normal vibrations
M1 – With Load	M1	With load	6	L2	66%	0.40	3.00	~Vtech L3 level
M1 – With Load	M1	With load	6	L3	100%	0.42	3.00	Too much vibration
M2 – With Load	M2	With load	6	L1	33%	0.36	2.16	Small motor, good start
M2 – With Load	M2	With load	6	L2	66%	0.66	3.96	Good torque
M2 – With Load	M2	With load	6	L3	100%	1.10	6.60	High vibrations
M3 – N/A	M3	N/A	—	—	—	—	—	Too slow with noisy gearbox / too fast without

5.3 Key Findings

- M8 at L2 (66% PWM, 6V) produced vibration closely matching Vtech L3 — selected as primary candidate.
 - M7 was underpowered at all levels — too slow even at 100% duty cycle. Eliminated.
 - M3 eliminated due to gearbox noise and inconsistent speed.
 - M1 and M2 showed comparable performance to M8 but slightly lower power draw.
 - Vtech motor operates best at 4 V (2 W). Board voltage range L1–L3 (4.5–6 V) confirmed via measurement.
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6. Vtech V-Hush Rocker — Reverse Engineering

6.1 Disassembly

Vtech V-Hush Baby Rocker was fully disassembled. PCB photographed, all ICs and passive components identified by marking and datasheet lookup.

6.2 Bill of Materials

#	Component	Description
1	Speaker	8 Ω , 0.5 W
2	Night Light LEDs	3.3 V forward voltage, LCSC C965810
3	Transistor	MMBT3904 1AM — LED driver
4	Li-ion Charger IC	ME4057ASPG (4057A)
5	Memory IC	Winbond 25WQ16JV EEPROM, 2 MB
6	Boost IC	LP6460 — 3.6–4.2 V $\rightarrow$ 5 V, 40 mA
7	Motor	JC25A-370-10.5K, 6 V DC
8	Speaker & MCU IC	AC24bp27986 42a4

6.3 Power Architecture Findings

- Single Li-ion cell (3.6–4.2 V) is the supply source.
- LP6460 boost IC steps up to stable 5 V for MCU and peripherals. Very low quiescent current — 40 mA rated.
- Motor (JC25A-370-10.5K) runs directly from board supply levels L1–L3 via PWM.
- ME4057A handles Li-ion charging from USB input.
- Winbond 2 MB EEPROM used for audio file storage.

6.4 Individual Component Testing

- Speaker (8 Ω , 0.5 W): tested with function generator — confirmed working.
- Night light LEDs (C965810): confirmed 3.3 V forward voltage, driven via MMBT3904 transistor.
- ME4057A charger IC: Li-ion charging cycle confirmed with bench cell.
- LP6460 boost: output confirmed stable at 5.0 V across load range.
- Winbond EEPROM: SPI communication confirmed.

7. Vibration Mechanism Study

7.1 ERM — Eccentric Rotating Mass

Eccentric mass principle selected as primary soothing mechanism, consistent with Vtech implementation. An offset weight attached to the motor shaft generates centrifugal imbalance as it rotates, producing vibration in the mounting surface.

Parameter	Notes
Frequency range	5–20 Hz for baby soothing. Sweet spot 10–15 Hz
Intensity control	Motor PWM duty cycle — higher speed = stronger vibration
Weight offset	Greater offset = stronger vibration at same RPM
Limitation	Frequency and amplitude are coupled — cannot change independently

7.2 Coin and PCB Vibration Motor Tests

- Tested standard coin vibration motor (3 V rated): good vibration feel, very low torque.
- Tested small PCB vibration motor: acceptable for lightweight applications.
- Both eliminated in favour of larger shaft-mount DC motors for better intensity control.

7.3 CAD Design

- Designed eccentric weight mount in CAD matched to motor shaft diameter.
- Four weight variants designed with different offset distances to tune vibration intensity in progresss.
- Mechanism CAD finalised for 3D printing and bench-mount testing.

8. MCU Evaluation & Selection

8.1 Candidates Evaluated

Feature	RP2040	STM32F103	ESP32-S3-N16R8
Native USB MSC	Yes	No	Yes
BLE	No (external)	No (external)	BLE 5.0 native
WiFi	No	No	2.4 GHz native
Flash storage	2 MB (external)	64 KB	16 MB (on-chip)
OTA update	No	No	Yes
RTC	No	Basic	Built-in

Verdict	Partial fit	Eliminated	Selected
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8.2 Selection Rationale

ESP32-S3-N16R8 selected as the production MCU. Key reasons:

- Native BLE 5.0 eliminates external BLE module — reduces PCB size and component count.
- Native USB OTG supports USB MSC for drag-and-drop audio file management.
- 16 MB on-chip flash eliminates external flash or SD card for audio storage.
- WiFi enables OTA firmware updates and web-based file upload as alternative to USB MSC.
- Dual-core Xtensa LX7 with 8 MB PSRAM provides ample headroom for future features.
- STM32F103 eliminated — no USB peripheral, no wireless capability.

9. IMU Integration — MPU6050

9.1 Hardware

- MPU6050 module connected to ESP32-S3 via I2C (SDA/SCL).
- Mounted on motor test frame to measure actual surface vibration.

9.2 Firmware Development

- Wrote I2C read code for raw accelerometer registers (ACCEL_XOUT_H/L).
- Applied complementary filter and moving average to smooth raw readings.
- Tuned filter constants iteratively until sensitivity matched physical vibration perception.
- Serial output verified against hand-feel at each motor speed level.

9.3 Planned Use

- Real-time vibration feedback loop: compare measured vibration against target level.
- Adjust motor PWM automatically if vibration deviates from setpoint.
- Replace handheld vibration meter for motor comparison data collection.

9.4 Vibration Measurement — MPU6050 vs AS63B

Parameter	MPU6050 + Python	AS63B
Low frequency (5–10 Hz)	Excellent	Marginal (10 Hz lower limit)
FFT frequency output	Yes — Python scipy.fft	No — amplitude only
Data logging	Continuous CSV	Manual only

3-axis	Yes	Single axis
Closed loop use	Yes — embeddable	No
ISO certification	No	ISO 2954
Cost	~Rs. 350 (in BOM)	Rs. 29,500
Decision	Selected for project	Not purchased

10. Next Steps

#	Task	Target	Status
1	Fabricate and fit eccentric weights on all test motors	Week 1	Pending
2	Motor comparison test - 4 weights with same motor and with MPU6050 data recording	Week 1	Pending
3	Finalize motor + eccentric weight combination for product	Week 1	Pending
4	Implement 3 speed levels + button control on ESP32-S3	Week 2	Pending
5	Integrate audio module and test sound playback	Week 2	Pending
6	Full system integration and prototype assembly	Week 4	Pending